[INCH-POUND]
GGG-K-450C
April 5, 1996
SUPERSEDING
GGG-K-00450B
September 14, 1970

FEDERAL SPECIFICATION

KNIFE, CRAFTSMAN'S, MULTIPURPOSE

(STENCIL, ETCHING, AND CUTTING, REMOVABLE BLADES)

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers multipurpose knives having chucks for holding blades in the fixed or swivel position.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. The knives shall be of the following types, classes, and styles, as specified (see 6.1):

Type I - Fixed blade

Class 1 - Light-duty holder (fixed)
- Style 1 - Curved blade
- Style 2 - Beveled blade
- Style 3 - Stencil cutting blade
- Style 4 - Square blade

Class 2 - Medium-duty holder (fixed)
- Style 1 - Broad blade
- Style 2 - Curved blade
- Style 3 - Double-edged blade
- Style 4 - Close-corner blade

Type II - Swivel blade

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5110

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40	-	Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-151	-	Metals Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

American Society for Testing and Materials (ASTM):

ASTM E 18	-	Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of knives otherwise conforming to this specification.

3.2 Materials. Materials used in craftman's knives shall be of the best quality material as hereinafter specified.

3.3 Threaded parts. Threaded parts shall be in conformance with the applicable requirements of FED-STD-H28.

3.4 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.5 Finish.

3.5.1 Metal. Craftman's knives shall be smoothly finished in accordance with good commercial practices.

3.6 Type I, fixed blade. Type I knives shall consist of a handle, blade chuck, and blade(s).

3.6.1 Handle. The handle, including the blade chuck, shall be not less than 4-1/2 inches in length. The handle shall be aluminum, plastic, or other material having equal strength and durability, and be capable of passing the strength test specified in 4.5.4. The cross-sectional area of aluminum handles shall be not less than 0.076 square inch for class 1, and 0.150 square inch for class 2. The cross-sectional area of plastic handles shall be not less than 0.196 square inch having a flame resistance sufficient to withstand the test specified in 4.5.2.1.

3.6.2 Blade chuck. The blade chuck shall be made of either a corrosion-resistant metal, or a suitable metal with a corrosion-resistant finish. It shall be accurately machined and adjustable so that the chuck can be tightened to hold the blades securely in a fixed working position (see 4.5.1). The blade chuck shall be rigid and attached to the handle, and shall be knurled or serated to facilitate blade changes.

3.6.3 Blades. Blades ordered either separately or with the knife shall be of the same quality. Replacement blades shall be furnished with the manufacturer's regular shank, and shall be interchangeable in the blade chuck of the model number of knife specified (see 6.1). All blades shall be honed to a razor sharp edge and have a hardness of 58 to 63 on the Rockwell C scale (see 4.5.3).

3.7 Type I, class 1, light duty holder (fixed). Type I, class 1, blade chuck holders shall readily accept each style of blade specified for class 1 holders. Chuck holders shall be designed for light duty cutting and quick, easy blade insertion and removal. Unless otherwise specified (see 3.9), type I, class 1, holders shall be furnished with one style of blade, as specified (see 6.1).

3.7.1 Blades for type I, class 1, holder. Blades for type I, class 1, holders shall be not less than 3/4 inch, nor more than 1-1/4 inches in length exclusive of the shank. Blades shall be not less than 0.020 inch in thickness and blade shanks shall be not less than 0.250 inch nor more than 0.285 inch in width. Blades shall be similar in shape to Figure 1A, 1B, 1C, or 1D. The blades shall be packed in a rigid plastic container and the container shall be blister packed to a cardboard backing.

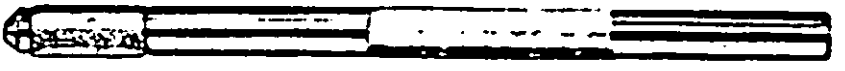


Figure 1. Type I, class 1, light duty holder (fixed).

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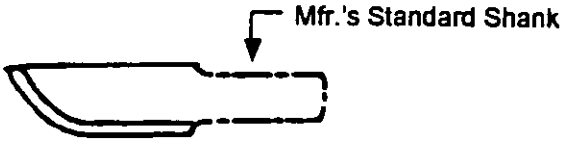


Figure 1A. Style 1, curved blade.

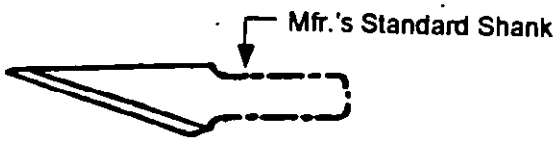


Figure 1B. Style 2, beveled blade.

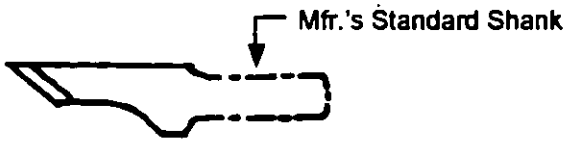


Figure 1C. Style 3, stencil cutting blade.

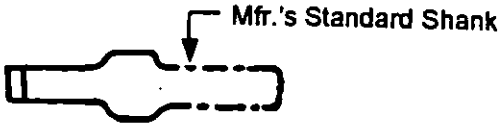


Figure 1D. Style 4, square blade.

3.7.1.1 Style 1, curved. Style 1 blades shall have a straight cutting edge, blending into a curved portion that terminates in a sharp point. The curved point shall have a radius of not less than 5/16 inch, and the straight portion shall be in a plane that is parallel to the horizontal axis of the knife handle (see Figure 1A).

3.7.1.2 Style 2, beveled. Style 2 blades shall have a straight cutting edge that is on a bevel (or angle) of not less than 20° nor more than 30° to the horizontal axis of the knife handle (see Figure 1B).

3.7.1.3 Style 3, stencil cutting. Style 3 stencil cutting blades shall have a straight cutting edge that is on an angle of not less than 40° nor more than 50° to the horizontal axis of the knife handle. The cutting edge shall terminate in a sharp point (see Figure 1C). Unless otherwise specified (see 6.1), the width of the straight narrow portion of the blade shall be not less than 1/8 inch nor more than 1/4 inch.

3.7.1.4 Style 4, square. Style 4 blades shall have a straight cutting edge that is square (90°) to the horizontal axis of the knife handle (see Figure 1D). Unless otherwise specified (see 6.1), the cutting edge length shall be not less than 1/8 inch nor more than 1/4 inch. The cutting edge length shall be approximately equidistant about the centerline of the handle.

3.8 Type I, class 2, medium duty holder (fixed). Type I, class 2, blade chuck holders shall readily accept each style of blade specified for class 2 holders. Chuck holders shall be designed for medium duty cutting and quick, easy blade insertion and removal. Unless otherwise specified (see 3.9), type I, class 2, holders shall be furnished with one style of blade, as specified (see 6.1).

3.8.1 Blades for type I, class 2, holders. Blades for type I, class 2, holders shall be not less than 1-3/8 inches nor more than 2-1/8 inches in overall length (including the shank). Blades shall be not less than 0.020 inch in thickness and blade shanks shall be not less than 0.310 inch in width. Blades shall be similar to shape to Figure 2A, 2B, 2C, or 2D. The blades shall be packed in a rigid plastic container and the container shall be blister packed to a cardboard backing.

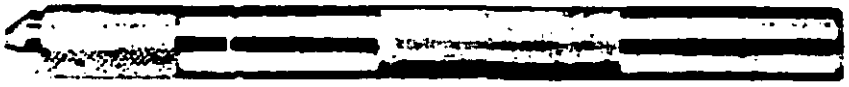


Figure 2. Type I, class 2, medium duty holder (fixed).

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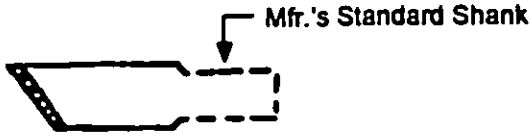


Figure 2A. Style 1, broad blade.

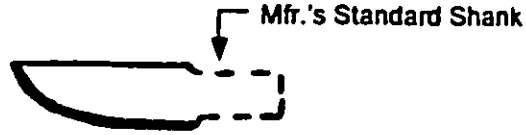


Figure 2B. Style 2, curved blade.

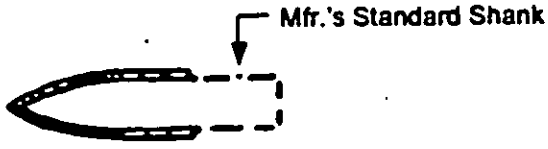


Figure 2C. Style 3, double-edged blade.

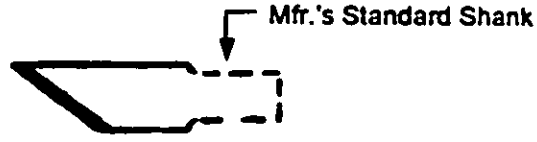


Figure 2D. Style 4, close-corner blade.

3.8.1.1 Style 1, broad blade. Style 1 blades shall have a straight cutting edge that is on a bevel (or angle) of not less than 40° nor more than 55° to the horizontal axis of the knife handle. The cutting edge shall terminate in a sharp point (see Figure 2A).

3.8.1.2 Style 2, curved blade. Style 2 blades shall have a straight cutting edge, blending into a curved portion that terminates in a sharp point. The curved portion shall have a radius of not less than 5/16 inch, and the straight portion shall be in a plane parallel to the horizontal axis of the knife handle (see Figure 2B).

3.8.1.3 Style 3, double-edged blade. Style 3 blades shall have two cutting edges parallel to the horizontal axis of the knife axis which blend into curved cutting edges and terminate in a sharp point at the center of the blade (see Figure 2C).

3.8.1.4 Style 4, close-corner blade. Style 4 blades shall have a straight cutting edge that is on a bevel (or angle) of not less than 25° nor more than 40° to the horizontal axis of the knife handle. The cutting edge shall terminate in a sharp point (see Figure 2D).

3.9 Sets. When ordered as a set (see 6.1), the set shall consist of the class of knife holder and at least two different styles of blades specified herein. The set shall include an additional holder for heavy duty application. Minimum overall length of holder (excluding blade) shall be 4-7/16 inches; plastic handle with 4 or 6 flutes; minimum length 2-7/32 inches; minimum diameter of handle 27/32 inch. The plastic handle shall pass the flame resistance test as specified in 3.6.1. Holder shall be furnished complete with one type I, class 2, style 1 broad blade. Holder shall be similar to Figure 3. The set shall be enclosed in a sturdy compartmented wood box plainly marked with the manufacturer's set number and name or identifying symbol.

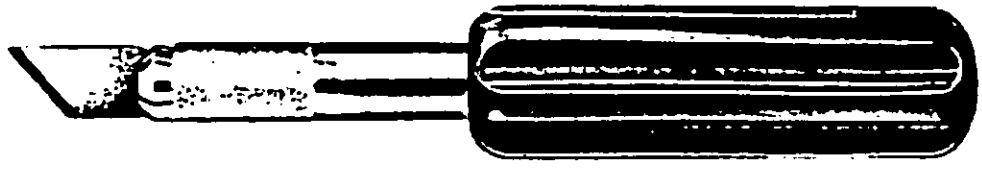


Figure 3. Heavy duty holder.

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3.10 Type II, swivel blade. Type II stencil knives shall consist of a handle with a blade chuck and a detachable blade on a swiveling shaft.

3.10.1 Handle. The handle, including the blade chuck, shall be not less than 4-1/2 inches in length and not less than 3/8 inch in diameter at the chuck end.

3.10.2 Blade chuck. The blade chuck shall be of suitable metal, accurately machined, and securely attached to the handle. Chucks shall be designed to permit the blade to either swivel when following a curved line, or be locked in a fixed position. Holder shall be similar to Figure 3. Chucks shall be knurled or serrated to permit easy hand adjustment.

3.10.3 Blade. The blade shall have a cutting edge at an angle of 45° ± 2 ° with the axis of the handle, and securely welded or brazed to the shaft. The blade shall be 1/4 inch ± 1/16 inch in width, and 7/16 inch ± 1/16 inch in length, and shall be in accordance with 3.6.3 in all other respects.

3.10.4 Type II knives. Type II knives shall be similar to Figures 4 or 5.

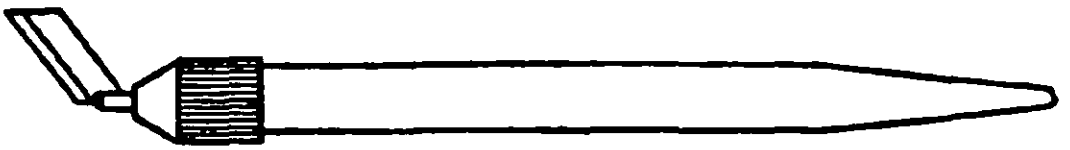


Figure 4. Type II, swivel-blade stencil knife.

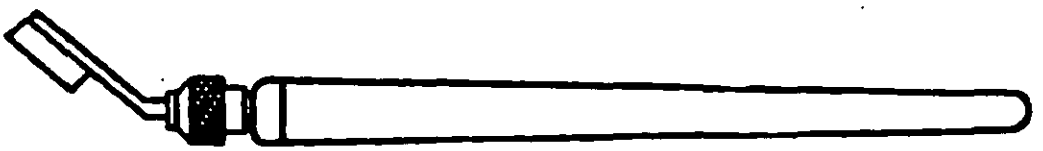


Figure 5. Type II, swivel-blade stencil knife.

3.11 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.12 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4. Data for sampling shall be as specified in Table I.

TABLE I. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level (AQL)	AQL expressed in terms of	Reference
Visual examination	1 each	II	1.5	Defects per hundred units	4.3.1
Dimensional examination	1 each	S-4	1.5	Defects per hundred units	4.3.2
Testing	1 each	S-3	1.0	Percent defective	4.5
Preparation for delivery	one container	S-2	4.0	Defects per hundred units	4.6

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. For packaging and packing defects see PPP-P-40.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.5.

4.5 Test procedures.

4.5.1 Practical cutting tests. Each of the sample knives selected shall be subjected to a practical cutting test by whittling and carving a small piece of lumber, such as a stick or kiln-dried soft pine. All type I, class 1 , style 3; type I, class 2, style 4; and type II blades shall also be subjected to cutting cloth and paper commonly used for mask and stencils. If any test sample blade fails to cut freely and cleanly, or if any blade loosens of slides in the chuck after being tightened for use or comes out of the blade chuck during the test, the lot shall be rejected.

4.5.2 Sampling for flammability tests of plastic handle knives. When applicable, a random sample of three plastic handle knives shall be taken from each lot and subjected to the flammability tests, except when lots consist of less than 40 knives; then the test may be omitted at the discretion of the inspector.

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4.5.2.1 Flammability test. The knife for test shall be clamped in a support with the handle extended outward in a horizontal position. A Bunsen burner, alcohol lamp, or match flame 1/2 to 3/4 inch in height shall be applied so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds, the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handle after the removal of the flame. In case the handle does not continue to burn after the first ignition, the flame shall be placed under the free end of the handle for a period of 30 seconds immediately following the extinction of the flame. If the handle does not continue to burn, or if it does not burn at a rate exceeding 2 inches per minute, the plastic handle meets the requirements of this test.

4.5.3 Hardness test. Samples of each type and class of knife shall be tested in accordance with ASTM E 18 for determination of compliance with hardness requirements specified in section 3.

4.5.4 Strength test. The assembled unit, knife and blade, shall be placed vertically in a compression testing machine with the blade point on a block of hardwood or fiberboard. A gradually increasing downward force shall be applied until a maximum of 30 pounds has been reached. After this force has been removed, there shall be no visual evidence of bend or broken blades, or slippage of the blade in the chuck. Failure to meet this test shall be cause for rejection.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class of holder and style of blade required (see 1.2.1 and 3.7 or 3.8).
c. When replacement blades are required, specify style of blade and the manufacturer's name and model number of knife (see 3.6.3).
d. Width of class 1, style 3 blade, if different (see 3.7.1.3).
e. Cutting edge length of class 1, style 4 blades, if different (see 3.7.1.4).
f. If set required, specify class of knife and styles of blades required (see 3.9).
g. Level of preservation, packaging, and packing required (see 5.1).
h. Marking, if other than specified.

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